The Ramification Index:

Version 4: A Fourth Firm, and Why It Fails the Scope Condition

Khayyam Wakil
The ARC Institute of Knowware
Calgary, AB
the@knowware.institute

August 2026

Abstract

This fourth version continues the audit line of Version 3 (*Provenance, Construction, and What the Series Can Support*), not the expansion line of the bifurcated-index paper. It asks a single question: does the most widely reported structural event in the DRAM market since the series began—the July 2026 public listing of a fourth manufacturer, ChangXin Memory Technologies (CXMT)—change what the series can support? It concludes that it does not, and that the honest effect of CXMT is to *tighten* the scope condition rather than to extend the index. Four points. (1) CXMT's price is policy-administered: a state self-sufficiency mandate, export-control isolation, a blacklisted buyer set, and an equity that opened up 466% pricing a geopolitical option rather than cash flow. By the scope condition of Version 3 it lacks a fast, elastic, private-market price and belongs with potash—untestable—not with incumbent DRAM. (2) The nearest precedent to 2025–2026 is the 1988 shortage, which Version 3 showed is missing from the published series; both episodes are trade-policy-driven (1988 followed the 1985 US–Japan Semiconductor Agreement; 2026 follows export controls and state subsidy). This strengthens the withdrawal of the "without historical precedent" claim rather than motivating a new regime, and it makes any "CXMT regime" doubly untestable, since one of its two instances is absent from the data. (3) A fourth firm optimising for self-sufficiency rather than margin is a mid-sample break in the data-generating process; it makes the pooled $n = 46$ panel that Version 3 found underpowered *less* poolable, not more. (4) The geographic split (China-domestic versus rest-of-world) is populated for zero realised observations and is logged as a descriptive watch-item, exactly as Version 3 demoted the commodity/AI sub-indices. No new inferential claim is made. One claim from an earlier draft of this version—that a subsidised fourth entrant constitutes a fifth interpretive regime—is withdrawn here as premature.

1 Introduction

Version 3 of this paper exists in two documents that should not be confused. The first extends Version 2 with a practitioner-gap section and retains its confident register. The second—the one this version continues—applied the companion study's eight-guard protocol to the paper's own claims and withdrew most of them: the "without historical precedent" claim, the McCallum–TrendForce chain-link description, and any inferential reading of the bifurcated sub-indices, with the headline Granger result restated as untested rather than established [Wakil 2026c]. The controlling standard, in other words, is not "what can we add?" but "what can the series carry?"

An earlier draft of *this* version failed that standard directly. Presented with the CXMT listing, it proposed a fifth interpretive regime ("subsidised-supply divergence") and a second, geographic

bifurcation of the index (R_t^{dom} against R_t^{RoW}), and it built a figure for each. Both are withdrawn below, for the same reason Version 3 withdrew the commodity/AI sub-indices: a regime you cannot yet measure is a narrative, not a finding, and dressing a news item in subscripts does not make it an estimate. The present version reports what CXMT does to the scope condition, and nothing it has not run.

2 The event, and why the equity is not a price

CXMT listed on the Shanghai STAR Market on 27 July 2026, pricing 6.688 billion shares at ¥8.66 to raise ¥57.92 billion (about US$8.6 billion), and rose roughly 466% on debut to a market capitalisation near ¥3.3 trillion (about US$488 billion), briefly the largest of any mainland-listed company. Independent trackers place it fourth in global DRAM at roughly 8% of 2025 shipments, behind Samsung (36%), SK hynix (29%), and Micron (24%); its own prospectus reports 7.67% on a Q4-2025 revenue basis.

None of this is a price signal of the kind the index reads. At the offer price the trailing P/E was near 309× against a forward P/E near 5.8×: the debut priced the strategic value of a domestic champion, not a cash-flow forecast. An equity that moves 466% in a session on a policy expectation is a barometer of export-control risk, and it is treated here as one—tracked alongside the index as a policy variable, never inside it.

3 The scope condition, applied to CXMT

Version 3, Section 10, converted the Ramification Index from a working example into a class of indicator with a falsifiable scope condition: *concentrated buyers, discretionary forward commitment, and a fast, elastic, private-market price.* Agriculture fails the first two; potash satisfies concentration but its cartel-administered price fails the third and is marked untestable. CXMT is the instructive case because it *passes the first two and fails the third.*

Scope criterion	**CXMT status**
Concentrated buyers	Yes — the same few hundred capacity-planning firms buy from CXMT as from the incumbents.
Discretionary forward commitment	Yes — DRAM is bought 18–24 months ahead of deployment, CXMT included.
Fast, elastic, private-market price	**No** — price is shaped by state subsidy, a self-sufficiency mandate, export controls, and a buyer set constrained by a defence blacklist.
Verdict	**Untestable.** In scope by structure, out of scope by price — the potash case.

Table 1: CXMT against the Version 3 scope condition. Passing two of three criteria is not passing; the third is the one that made potash untestable.

The consequence is not that CXMT corrupts the existing series. The published index is built on incumbent-dominated ASPs (TrendForce, McCallum), and those remain what they were. The consequence is prospective and precise: a fourth source cannot be admitted, and no China-domestic

sub-index can be built, without first demonstrating that the price being admitted is a market price rather than an administered one—the demonstration potash never passed. To the extent CXMT's share grows and its administered price begins to move the *global* ASP the index reads, it injects an administered component into a previously clean signal. That is a contamination risk to log, not a regime to name. Figure 1 places CXMT on the Version 3 scope diagram.

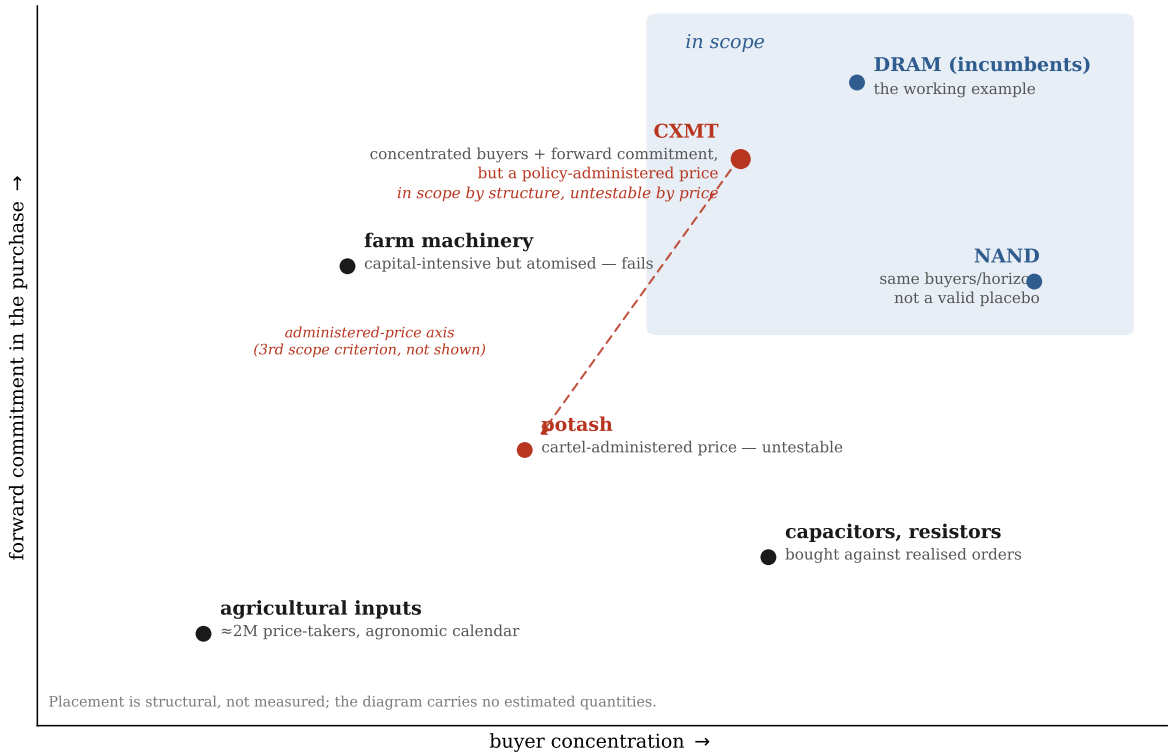


Figure 1: The scope condition with CXMT added. On buyer concentration and forward commitment CXMT plots inside the in-scope region, beside incumbent DRAM. It is coloured as untestable because it fails the third scope criterion—a fast, elastic, private-market price—which the two plotted axes do not capture. The dashed arrow marks the axis it shares with potash. Placement is structural, not measured.

4 Precedent: 1988 was also trade policy

Version 3 withdrew Version 2's claim that 2025–2026 is "without historical precedent" in the 46-year series, on the ground that the series omits the 1988 shortage—a supply-driven spike of +82.5% in the recovered archive against −21.8% in the published table. The 1988 episode followed the 1985 US–Japan Semiconductor Agreement: a trade intervention, then capacity underinvestment, then a shortage. The 2025–2026 episode is export controls plus a state-subsidised entrant: also a trade intervention.

So the two largest supply episodes in the series' conceptual history are both trade-policy-driven. This does not restore the uniqueness claim; it converts the phenomenon from "unprecedented shock" to "the second observation of a recurring political-economy mechanism." It also has a sharp methodological consequence. Any test of a "CXMT regime" or a "trade-policy supply regime"

would require both instances in-sample—and one of them, 1988, is exactly the observation Version 3 showed is missing from the data. The regime is therefore untestable twice over: once because CXMT's own price fails the scope condition, and once because its only historical comparator is not in the series.

5 A fourth firm is a mid-sample structural break

The central finding of Version 3 is not that $n = 46$ is small. It is that the annual Granger relationship is carried entirely by 2020–2026 and vanishes on the preceding forty observations, so the data cannot distinguish "DRAM became macro-relevant" from "two unusual episodes in a row." CXMT bears on that question in the wrong direction for anyone hoping to resolve it favourably.

A fourth producer optimising for national self-sufficiency rather than margin changes the mechanism the index is supposed to read—the oligopoly's capital-allocation discipline—partway through the sample. Whatever data accumulates after 2026 will be generated by a four-firm market with one non-profit-maximising participant; it is not drawn from the same process as the pre-2020 three-firm data. Pooling them estimates an average over two regimes and reports it as one. The licensed monthly ASP series that Version 3 identified as the precondition for testing the index must therefore also be regime-dated: a longer series that straddles the entry of a state-backed competitor does not buy the statistical power Version 3 asked for unless the break is modelled, and modelling it costs back the degrees of freedom the length was meant to supply.

Remark 5.1 *This is the same shape as Version 3's endpoint walk, one level up. There, the result moved as observations were withheld; here, the* generating structure *moves as the sample extends. Both are reasons the honest quantity to report is an interval of ignorance, not a point estimate.*

6 What is withdrawn, restated, and logged

Withdrawn (from the earlier draft of this version):

- The fifth interpretive regime ("subsidised-supply divergence") as a finding. It is a conjecture about a price movement that has not occurred, on a series that could not test it if it had.
- The geographic bifurcation $R_t^{\text{dom}}/R_t^{\text{RoW}}$ as inference. It is populated for zero realised observations. It survives only as a descriptive watch-item.
- The claim that a subsidised fourth entrant "loosens" oligopoly pricing. On present evidence CXMT has exercised pricing power, not surrendered it; the long-run claim is untestable and is not made.

Restated:

- The oligopoly premise. Version 2's "three firms, $\sim$95%" now carries an asterisk: four firms, one of them state-backed and priced outside the scope condition. This is a scope caveat on the existing index, not a new result, and it does not alter any published estimate.

Logged as watch-items (descriptive only, no inference): CXMT quarterly capacity additions; the China-domestic versus rest-of-world price gap, as a measure of market fragmentation; and CXMT's equity as a policy barometer—explicitly not an index input. In the companion repository these are added to the tracker flagged `administered / out-of-scope`, so that CXMT (688825.SS) is recorded without being folded into any tested series.

7 Conclusion

The exciting reading of CXMT is that a fourth firm has broken a three-firm oligopoly and the index needs new machinery to track it. The honest reading is the reverse. CXMT's price is administered and fails the scope condition; its only historical comparator is missing from the data; and its entry is a structural break that makes the underpowered panel harder to use, not easier. The correct response is the one Version 3 already prescribed—acquire regime-dated, scope-qualified monthly data—not to add a storey to a series whose foundations that version spent its length inspecting. A good protocol converts a headline into a caveat. Applied to CXMT, it does so again.

References

[1] Wakil, K. *The Ramification Index, Version 2: The Bifurcated Index, Granger Causality, and the 2025–2026 Supply-Side Divergence.* The ARC Institute of Knowware, 2026.

[2] Wakil, K. *The Ramification Index, Version 3: The Bifurcated Index, Granger Causality, Supply-Side Divergence, and the Practitioner Gap.* The ARC Institute of Knowware, 2026.

[3] Wakil, K. *The Ramification Index, Version 3: Provenance, Construction, and What the Series Can Support.* The ARC Institute of Knowware, 2026.

[4] Wakil, K. *Six Ways to Find a Leading Indicator That Isn't There.* Working paper, The ARC Institute of Knowware, 2026.

[5] Reuters; CNBC; Counterpoint Research; TechTimes; Fortune, coverage of the CXMT STAR Market listing, 27–28 July 2026. STAR ticker 688825 (Yahoo: 688825.SS). Facts verified against multiple outlets; conversions approximate.